Smart Dermatology Assistant using AI and Image Processing

ITA0506-Computer Vision

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INTRODUCTION

Skin diseases affect millions and require timely diagnosis.

AI-based image analysis improves accuracy and speed of detection.

The system identifies disease type, severity, and risk level automatically.

Users receive personalized care and treatment recommendations instantly.

OBJECTIVES

- To develop an AI-based system for automatic skin disease detection
- To analyze skin images using image processing techniques
- To accurately classify common skin diseases using CNN models
- To provide basic care, treatment, and preventive recommendations
- To assist in early diagnosis and support remote healthcare services

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Problem Statement

Skin disease diagnosis is mainly performed through manual visual inspection.

Diagnosis accuracy depends on the dermatologist's experience and expertise.

Manual examination is time-consuming and may delay treatment.

Early-stage skin diseases can be difficult to identify accurately.

Existing systems lack automated AI-based disease detection.

Severity of skin diseases is not assessed consistently.

People in remote areas have limited access to dermatologists.

There is a need for an AI-powered system to detect skin diseases from

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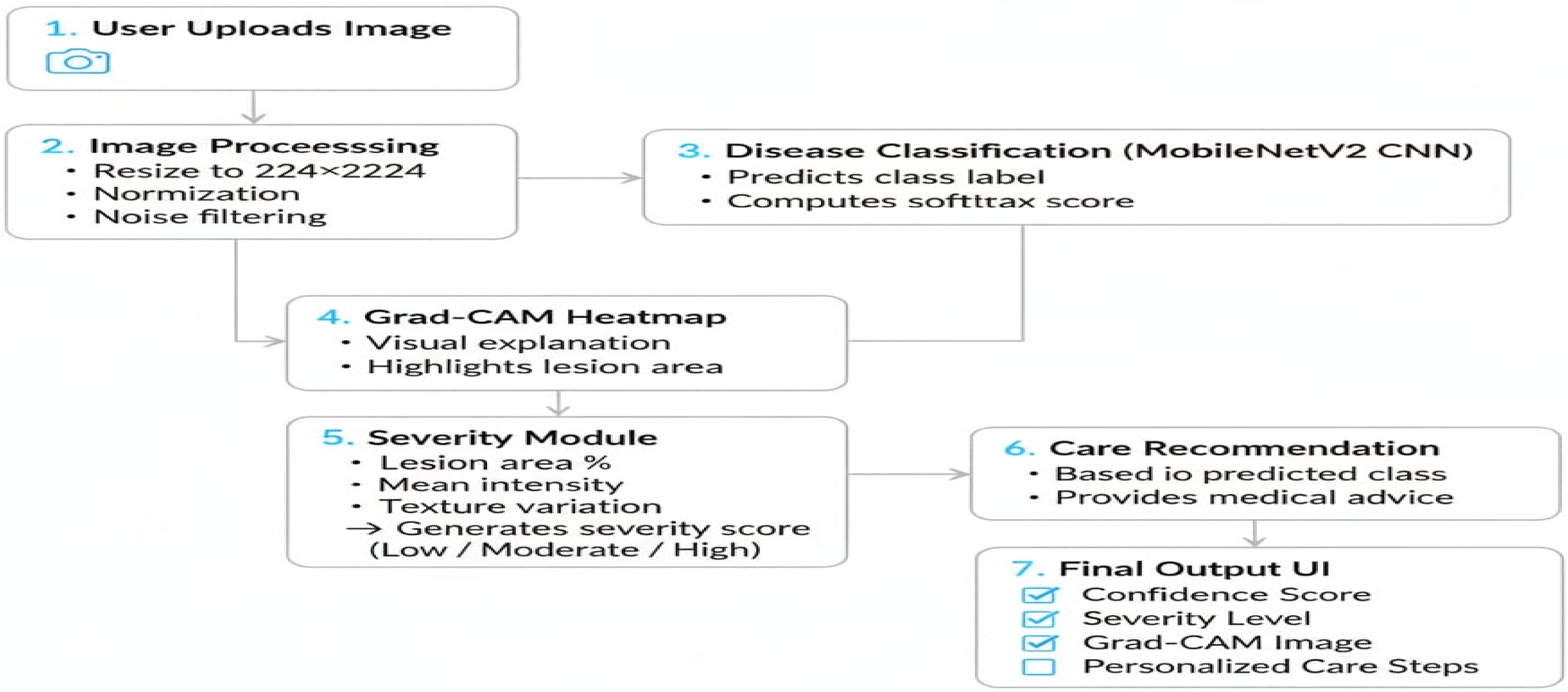
ARCHITECTUR DIAGRAM

AI Skin Lesion Diagnostic Assistant Architecture



LITERATURE SURVEY

Year	Author	Method used	Dataset	Outcome
2020	Wu et al.	CNN-based skin disease	Clinical skin	High accuracy in detecting
2021	Srinivas et al.	MobileNet (Deep	Kaggle skin	Lightweight model suitable
2022	Al-Masni	CNN + Segmentation	ISIC dataset	Improved accuracy using lesion
2023	Ahammed et al.	ML + Image	ISIC dataset	Skin disease detection
2024	Marri et al.	AI Mobile App (CNN)	Real-world user images	Effective for early detection approach awareness

Module 1: Skin Disease Detection using AI & Image Processing

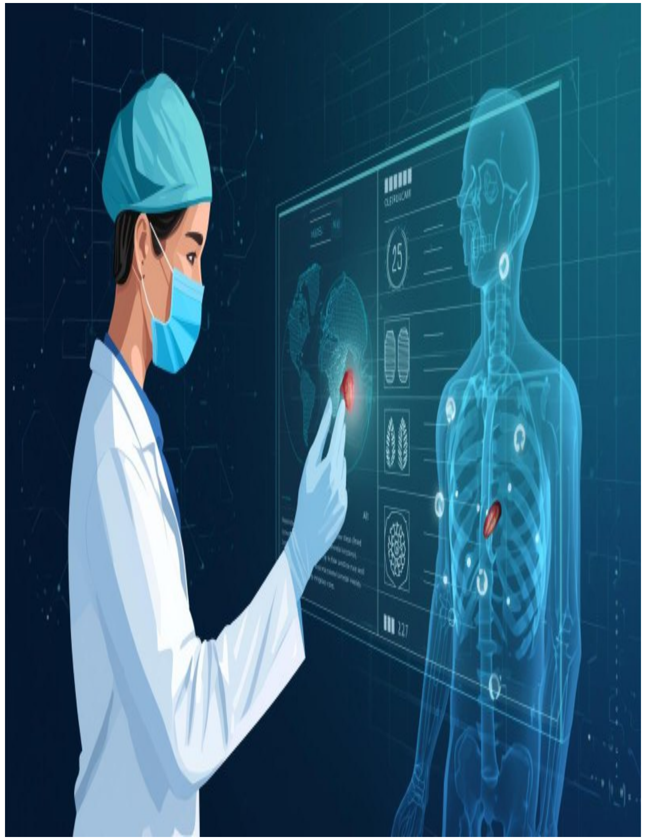
We collected different types of skin disease images for training the model.

The images were cleaned and resized using preprocessing techniques.

We used the MobileNetV2 deep learning model to detect skin diseases.

The model studies color, texture, and shape of the skin lesion.

It gives the disease name and prediction result for further



Module 2: Severity Analysis & Risk Level Prediction

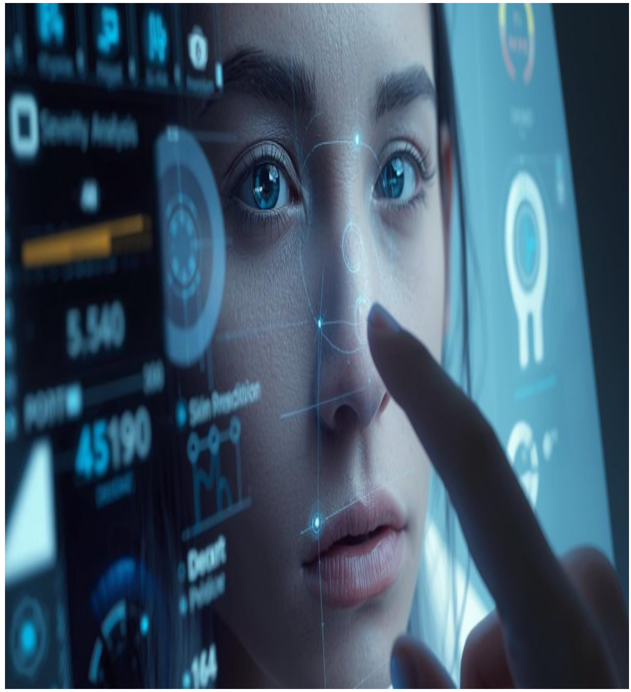
Extracted detailed lesion features such as area coverage, texture variation, and color intensity.

Computed a severity score using a weighted feature evaluation system.

Classified each case into Mild, Moderate, or Severe categories.

Designed a structured severity report for every input image.

Enabled risk-aware decision making by linking severity to the recommendation module.



Module 3: Care Recommendation & Preventive Guidance System

Built a rule-based system to suggest care based on disease type.

Gave medically accurate home-care tips and safety precautions.

Suggests doctor consultation when needed.

Recommended dermatologist visits for moderate and severe cases.

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Module 4: Explainable AI (Grad-CAM) Visualization

Applied Grad-CAM to visualize the regions of the skin image that influenced the AI model's prediction.

Generated heatmaps highlighting the affected lesion areas.

Improved transparency and reliability of the deep learning model.

Helped users and doctors understand why a particular disease was predicted.

Increased confidence in AI-assisted diagnosis.

Module 5: Report Generation & Patient History Management

Generated a comprehensive report containing disease name, confidence score, severity level, and care recommendations.

Stored patient details and prediction history for future reference.

Allowed comparison of previous and current skin conditions.

Supported monitoring of disease progression over time.

Enabled easy sharing of reports with healthcare professionals.

CONCLUSION

Smart system enables early and accurate detection of skin diseases

Uses AI and image processing to reduce manual diagnosis errors

Provides basic care, treatment, and prevention recommendations

Saves time and cost by reducing frequent hospital visits

FUTURE SCOPE

Improve accuracy using advanced deep learning and larger datasets

Support detection of a wider range of skin diseases, including rare conditions

Integrate with mobile apps and cloud platforms for real-time analysis

Enable direct online consultation with dermatologists

Add multilingual support and personalized treatment

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THANK YOU!